

EDA Report

Generated: 2026-04-08 18:03:40

Dataset Overview

- Data path: bank_aml_tm_synthetic_50k_local=data/bank_aml_tm_synthetic_50k_local.csv
- Rows used: 50000
- Target column: sar_label
- Time column: value_date

Data Quality

- Dataset has 50000 rows and 35 columns.
- Duplicate row ratio: 0.00%.
- Highest missing rate is 1.99% in 'merchant_category'.

Missingness

Column	Missing Count	Missing Rate
merchant_category	997	1.99%
counterparty_account_id	734	1.47%
device_type	629	1.26%

The following columns have no missing values: txn_id, txn_datetime, customer_id, account_id, txn_amount, currency, origin_country, destination_country, txn_channel, payment_rail (+22 more) (32 columns total).

Null-like / Placeholder Values

These values are not nulls but may represent missing or invalid data placeholders.

Column	Null-like Count	Null-like Rate	Example Values
merchant_category	339	0.68%	unknown
device_type	323	0.65%	unknown

Column Type Classification

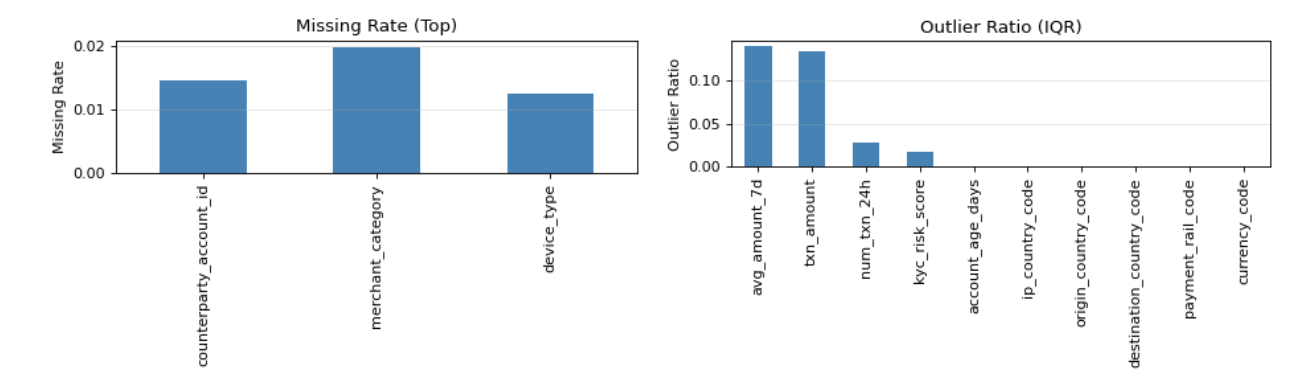
Type	Columns
numeric	txn_amount
numeric	account_age_days
numeric	kyc_risk_score
numeric	num_txn_24h
numeric	avg_amount_7d
numeric	origin_country_code
numeric	destination_country_code
numeric	ip_country_code
numeric	currency_code
numeric	txn_channel_code
numeric	payment_rail_code
numeric	txn_type_code

Type	Columns
numeric	merchant_category_code
numeric	device_type_code
categorical	currency
categorical	origin_country
categorical	destination_country
categorical	txn_channel
categorical	payment_rail
categorical	txn_type
categorical	merchant_category
categorical	device_type
categorical	ip_country
datetime	txn_datetime
datetime	value_date
boolean	is_pep
boolean	sanctions_match
boolean	is_cross_border
boolean	is_fintech_rail
boolean	sar_label

Outlier Ratio (IQR)

Column	Outlier Ratio
avg_amount_7d	0.140
txn_amount	0.134
num_txn_24h	0.028
kyc_risk_score	0.018
account_age_days	0
ip_country_code	0
origin_country_code	0
destination_country_code	0
payment_rail_code	0
currency_code	0

Charts



Target

- Target 'sar_label' has 2 classes.
- Target rate over time plotted.

Target Distribution

Value	Count	Rate
0	42,002	84.00%
1	7,998	16.00%

Target Rate by origin_country

origin_country	Target Rate
NG	25.43%
AE	24.98%
TR	22.76%
MA	22.61%
MX	21.71%
DE	15.60%
SG	15.54%
HK	15.46%
GB	15.20%
CH	15.06%

Target Rate by destination_country

destination_country	Target Rate
NG	36.13%
TR	33.49%
MX	32.74%
MA	32.32%
AE	32.14%
SG	8.45%
IN	8.30%
GB	8.09%
IT	7.92%
BR	7.44%

Target Rate by ip_country

ip_country	Target Rate
ES	17.32%
AE	17.03%
CN	16.72%
DE	16.63%
SG	16.45%
MA	16.38%
US	16.16%

ip_country	Target Rate
IN	16.08%
HK	15.97%
FR	15.74%

Target Rate by currency

currency	Target Rate
NGN	25.43%
AED	24.98%
TRY	22.76%
MAD	22.61%
MXN	21.71%
SGD	15.54%
HKD	15.46%
GBP	15.20%
CHF	15.06%
EUR	15.04%

Target Rate by payment_rail

payment_rail	Target Rate
swift	31.48%
wire	29.40%
zelle	24.11%
cashapp	23.86%
paypal	13.00%
google_pay	12.45%
stripe	12.16%
adyen	11.71%
apple_pay	9.40%
venmo	9.38%

Target Rate by merchant_category

merchant_category	Target Rate
luxury	19.35%
travel	18.56%
charity	16.09%
grocery	15.84%
electronics	15.69%
utilities	15.65%
restaurants	15.51%
gaming	15.11%
rideshare	15.10%
crypto	15.01%

Target Rate by txn_type

txn_type	Target Rate
international_remit	35.12%
crypto_exchange	33.40%
cash_withdrawal	21.30%
p2p_transfer	9.18%
bill_pay	7.81%
merchant_pay	7.63%
loan_payment	7.35%
cash_deposit	6.93%

Target Rate by txn_channel

txn_channel	Target Rate
api	17.61%
web	16.12%
mobile_app	15.77%
branch	15.31%
call_center	15.22%

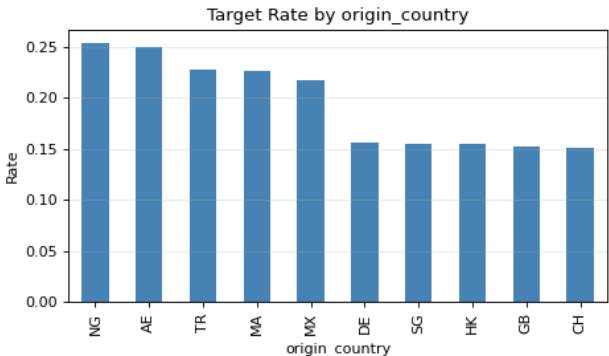
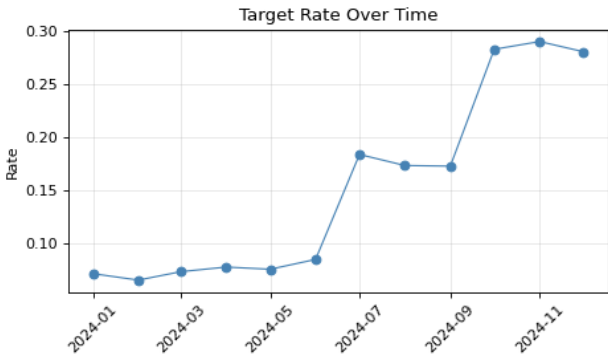
Target Rate by device_type

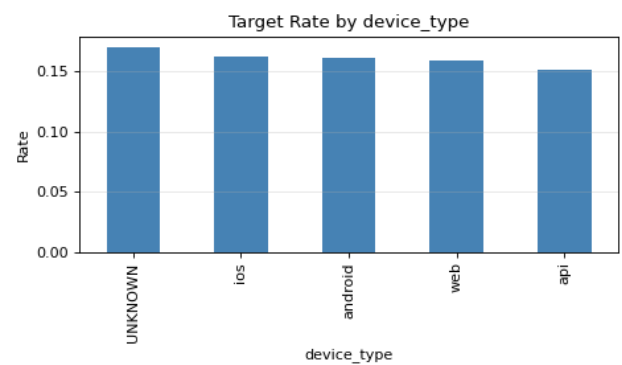
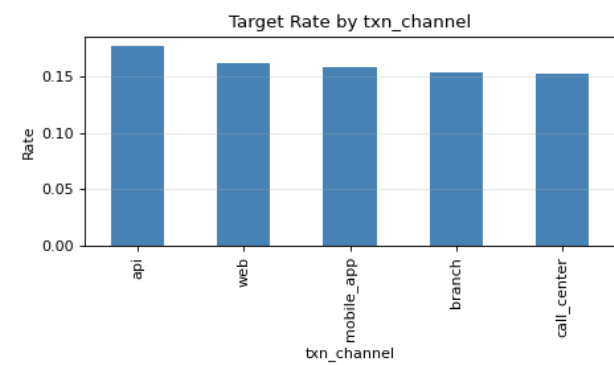
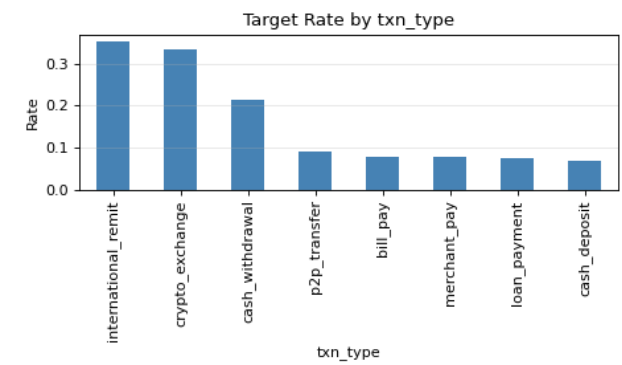
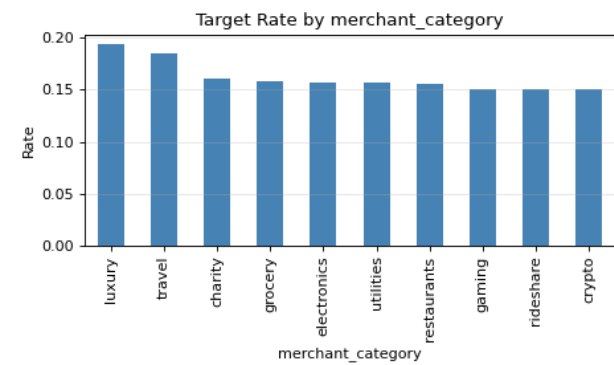
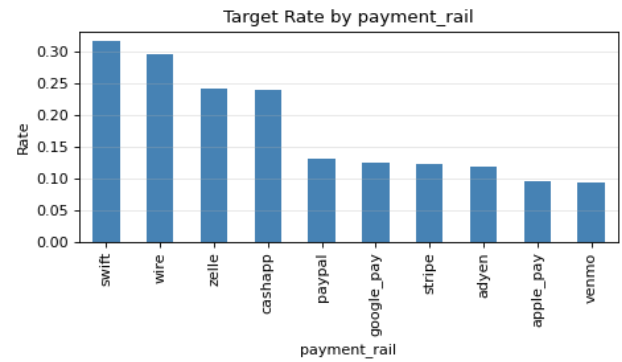
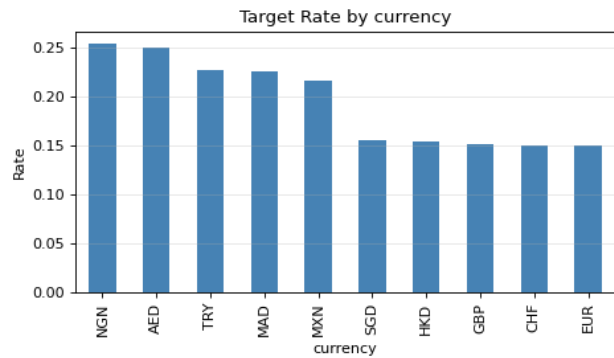
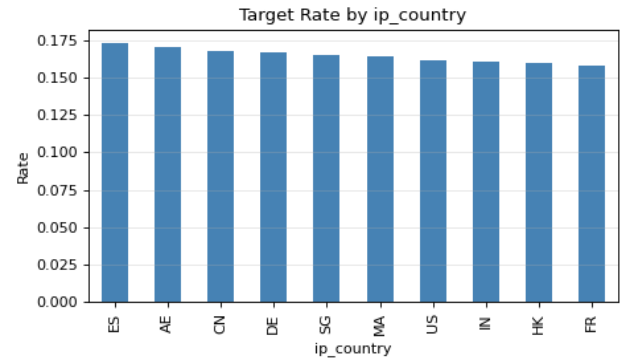
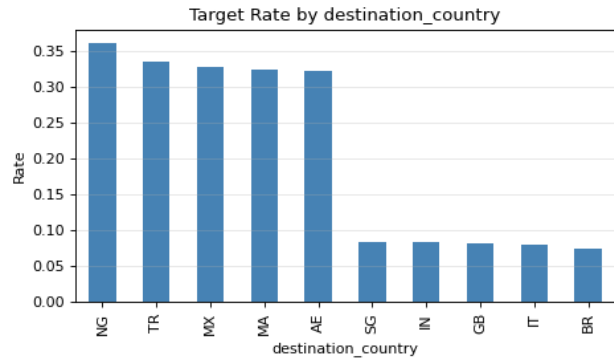
device_type	Target Rate
UNKNOWN	17.03%
ios	16.25%
android	16.10%
web	15.86%
api	15.12%

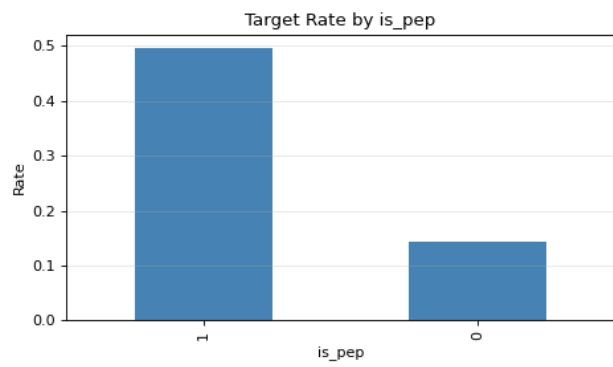
Target Rate by is_pep

is_pep	Target Rate
1	49.55%
0	14.43%

Charts







Univariate - Summary

Univariate

- Numeric columns analyzed: txn_amount, avg_amount_7d, account_age_days, kyc_risk_score, num_txn_24h, ip_country_code, origin_country_code, destination_country_code, payment_rail_code, currency_code
- Categorical columns analyzed: origin_country, destination_country, ip_country, currency, payment_rail, merchant_category, txn_type, txn_channel, device_type, is_pep

Numeric Summary Statistics

Column	count	mean	std	min	25%	50%	75%	max
txn_amount	50,000	1,347.008	17,824.971	0.440	46.778	127.275	423.370	1,826,089.880
avg_amount_7d	50,000	1,214.221	16,879.335	2.680	53.920	118.160	360.457	1,683,265.610
account_age_days	50,000	1,831.737	1,050.634	1	925	1,839.500	2,747	3,649
kyc_risk_score	50,000	39.880	18.147	0	27.530	38.671	50.722	100
num_txn_24h	50,000	4.249	7.759	0	1	3	5	89
ip_country_code	50,000	9.966	5.728	0	5	9	17	17
origin_country_code	50,000	9.991	5.726	0	5	10	17	17
destination_country_code	50,000	10.110	5.647	0	5	11	16	17
payment_rail_code	50,000	6.757	4.500	0	3	7	11	13
currency_code	50,000	7.206	4.335	0	4	5	13	13

Categorical Frequency (Top K)

Top K: origin_country

Category	Count	Rate
US	13,639	27.28%
DE	4,103	8.21%
GB	4,007	8.01%
FR	3,289	6.58%
ES	2,726	5.45%
IT	2,711	5.42%
BR	2,053	4.11%
CN	2,052	4.10%
IN	2,022	4.04%
NL	2,015	4.03%

Top K: destination_country

Category	Count	Rate
US	10,630	21.26%
MX	3,516	7.03%
MA	3,493	6.99%
TR	3,446	6.89%
AE	3,401	6.80%
DE	3,148	6.30%

Category	Count	Rate
GB	3,090	6.18%
NG	2,970	5.94%
FR	2,601	5.20%
IT	2,134	4.27%

Top K: ip_country

Category	Count	Rate
US	14,011	28.02%
DE	4,196	8.39%
GB	4,084	8.17%
FR	3,482	6.96%
ES	2,801	5.60%
IT	2,776	5.55%
BR	2,068	4.14%
CN	2,063	4.13%
NL	2,025	4.05%
IN	2,003	4.01%

Top K: currency

Category	Count	Rate
EUR	14,844	29.69%
USD	13,639	27.28%
GBP	4,007	8.01%
BRL	2,053	4.11%
CNY	2,052	4.10%
INR	2,022	4.04%
AED	1,669	3.34%
MAD	1,610	3.22%
TRY	1,582	3.16%
MXN	1,571	3.14%

Top K: payment_rail

Category	Count	Rate
wire	8,361	16.72%
ACH	8,321	16.64%
paypal	5,701	11.40%
swift	4,460	8.92%
card	4,100	8.20%
zelle	4,056	8.11%
cashapp	3,043	6.09%
rtp	2,735	5.47%
venmo	2,196	4.39%
sepa	2,076	4.15%

Top K: merchant_category

Category	Count	Rate
restaurants	8,867	17.73%
grocery	7,311	14.62%
utilities	6,839	13.68%
rideshare	5,861	11.72%
electronics	4,890	9.78%
travel	3,912	7.82%
gaming	3,871	7.74%
luxury	2,966	5.93%
charity	2,448	4.90%
crypto	2,038	4.08%

Top K: txn_type

Category	Count	Rate
p2p_transfer	11,956	23.91%
merchant_pay	8,243	16.49%
international_remit	7,386	14.77%
bill_pay	6,186	12.37%
crypto_exchange	5,155	10.31%
cash_withdrawal	4,807	9.61%
loan_payment	3,469	6.94%
cash_deposit	2,798	5.60%

Top K: txn_channel

Category	Count	Rate
mobile_app	22,345	44.69%
web	14,973	29.95%
api	5,274	10.55%
branch	4,958	9.92%
call_center	2,450	4.90%

Top K: device_type

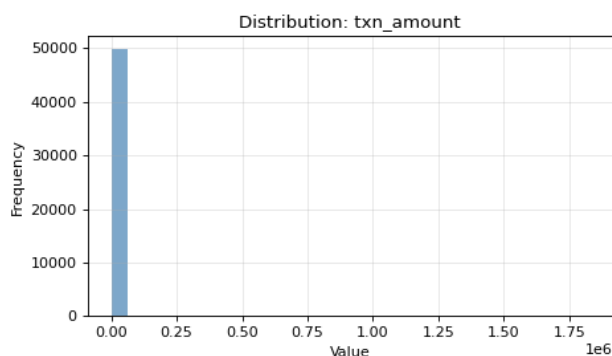
Category	Count	Rate
android	17,304	34.61%
ios	14,669	29.34%
web	12,386	24.77%
api	5,012	10.02%
UNKNOWN	323	0.65%
nan	306	0.61%

Top K: is_pep

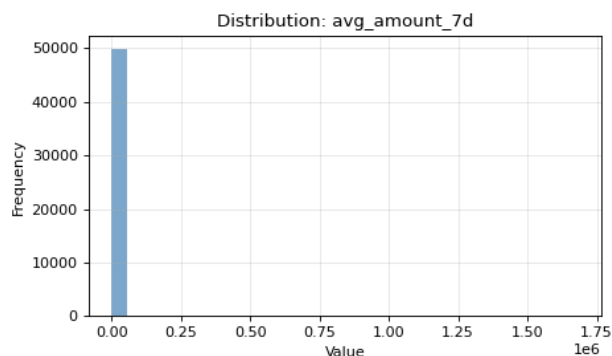
Category	Count	Rate
0	47,766	95.53%
1	2,234	4.47%

Univariate Plots

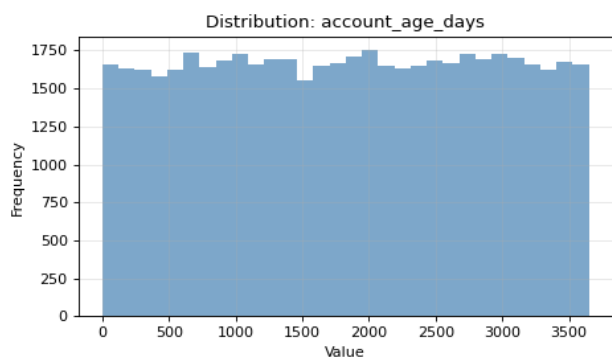
txn_amount distribution



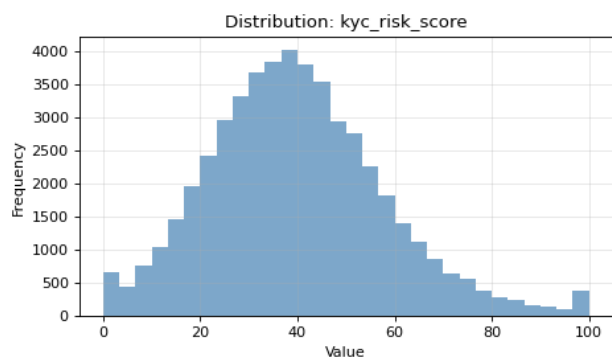
avg_amount_7d distribution



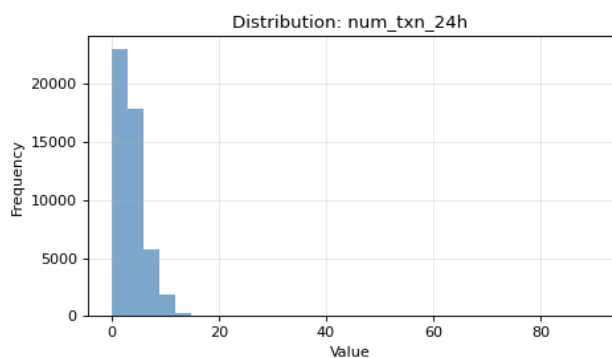
account_age_days distribution



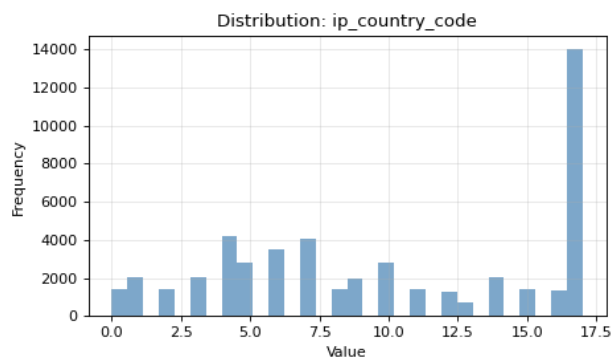
kyc_risk_score distribution



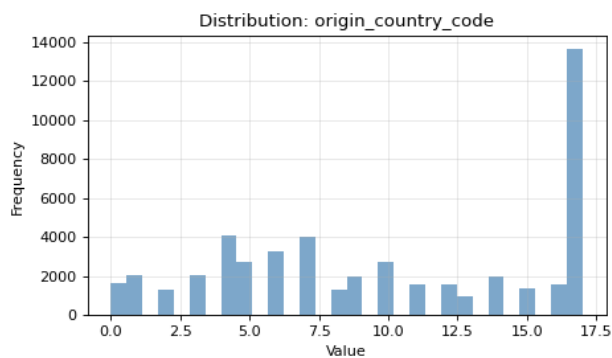
num_txn_24h distribution



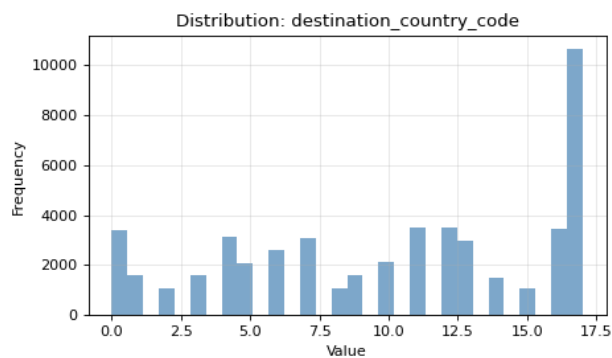
ip_country_code distribution



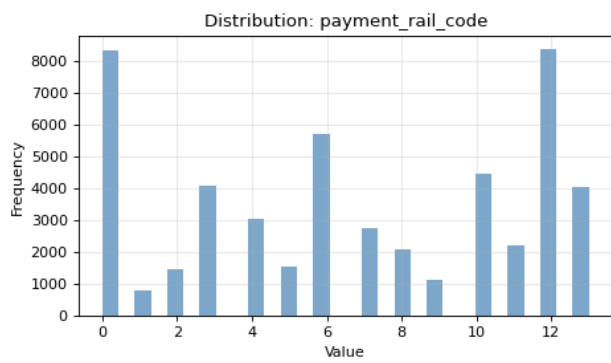
origin_country_code distribution



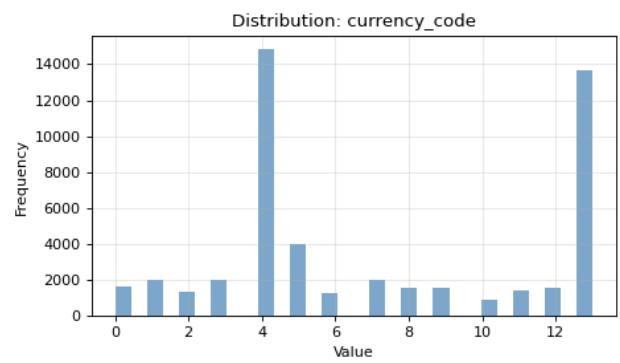
destination_country_code distribution



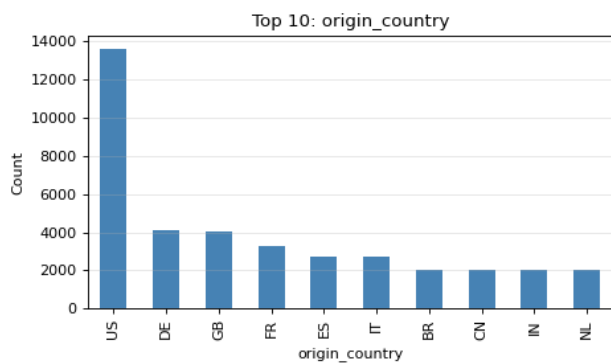
payment_rail_code distribution



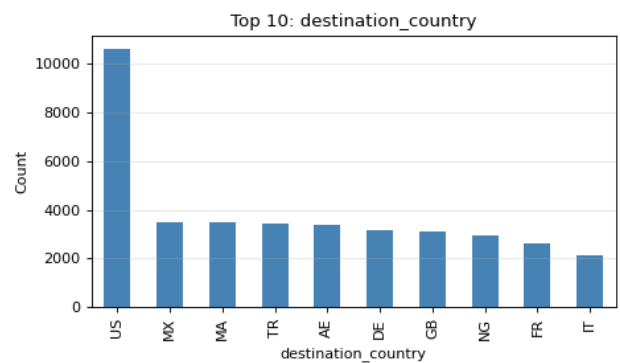
currency_code distribution



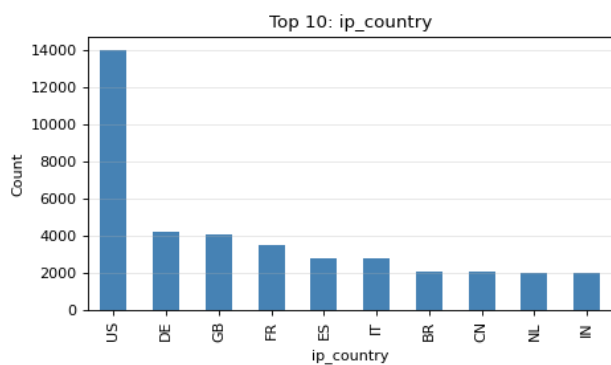
origin_country top 10



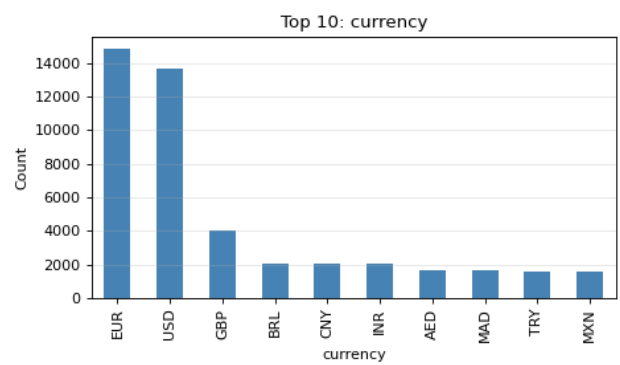
destination_country top 10



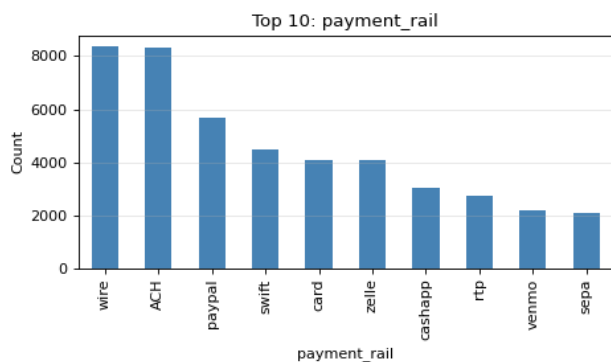
ip_country top 10



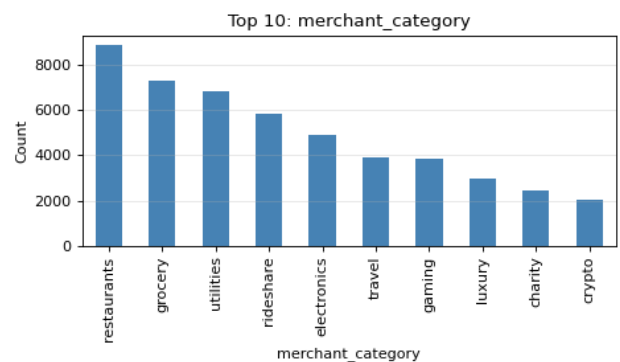
currency top 10



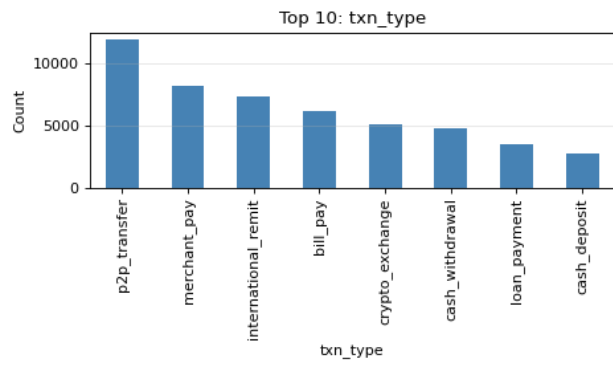
payment_rail top 10



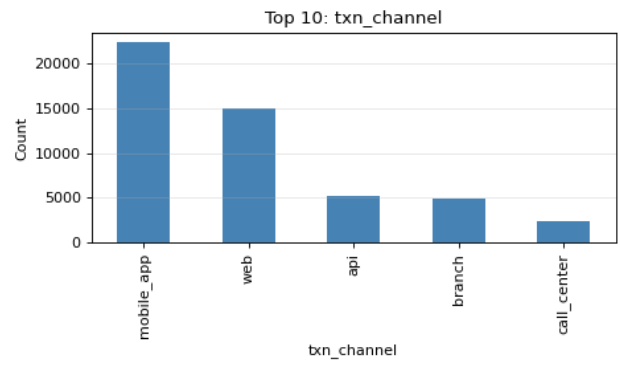
merchant_category top 10



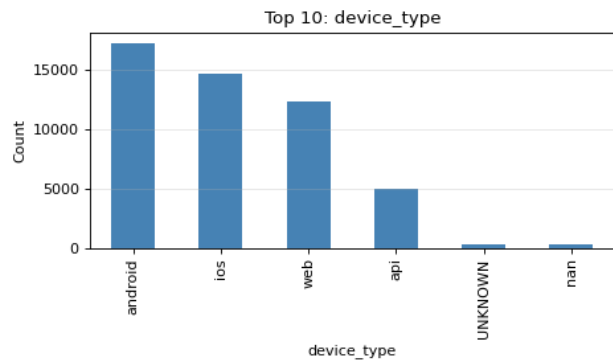
txn_type top 10



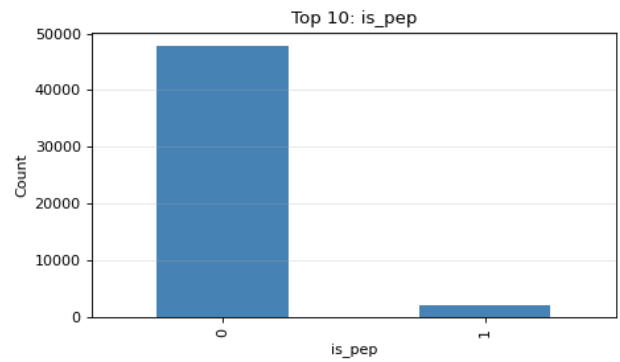
txn_channel top 10



device_type top 10



is_pep top 10



Bivariate Target

- Bivariate target analysis completed.

Numeric vs Target (Binned)

Column	Bin	Target Rate
txn_amount	(0.439, 36.788]	4.97%
txn_amount	(36.788, 85.22]	8.78%
txn_amount	(85.22, 195.502]	13.28%
txn_amount	(195.502, 583.078]	19.37%
txn_amount	(583.078, 1826089.88]	33.58%
avg_amount_7d	(2.6790000000000003, 45.5]	6.48%
avg_amount_7d	(45.5, 85.58]	8.96%
avg_amount_7d	(85.58, 171.808]	12.40%
avg_amount_7d	(171.808, 500.072]	19.33%
avg_amount_7d	(500.072, 1683265.61]	32.81%
account_age_days	(0.999, 741.0]	16.61%
account_age_days	(741.0, 1466.0]	15.96%
account_age_days	(1466.0, 2195.0]	15.32%
account_age_days	(2195.0, 2922.0]	16.30%
account_age_days	(2922.0, 3649.0]	15.79%
kyc_risk_score	(-0.001, 24.78]	6.48%
kyc_risk_score	(24.78, 34.48]	8.74%
kyc_risk_score	(34.48, 42.976]	12.45%
kyc_risk_score	(42.976, 53.87]	17.90%
kyc_risk_score	(53.87, 100.0]	34.42%
num_txn_24h	(-0.001, 1.0]	6.79%
num_txn_24h	(1.0, 2.0]	8.17%
num_txn_24h	(2.0, 3.0]	10.37%
num_txn_24h	(3.0, 5.0]	17.10%
num_txn_24h	(5.0, 89.0]	41.77%
ip_country_code	(-0.001, 4.0]	16.04%
ip_country_code	(4.0, 7.0]	16.09%
ip_country_code	(7.0, 12.0]	15.81%
ip_country_code	(12.0, 17.0]	16.01%
origin_country_code	(-0.001, 4.0]	16.43%
origin_country_code	(4.0, 7.0]	15.08%
origin_country_code	(7.0, 12.0]	17.35%
origin_country_code	(12.0, 17.0]	15.58%
destination_country_code	(-0.001, 4.0]	15.05%
destination_country_code	(4.0, 9.0]	7.56%
destination_country_code	(9.0, 12.0]	26.79%
destination_country_code	(12.0, 17.0]	15.97%
payment_rail_code	(-0.001, 2.0]	6.69%
payment_rail_code	(2.0, 6.0]	12.92%

Column	Bin	Target Rate
payment_rail_code	(6.0, 9.0]	6.48%
payment_rail_code	(9.0, 12.0]	27.09%
payment_rail_code	(12.0, 13.0]	24.11%
currency_code	(-0.001, 4.0]	15.65%
currency_code	(4.0, 8.0]	16.41%
currency_code	(8.0, 13.0]	16.20%

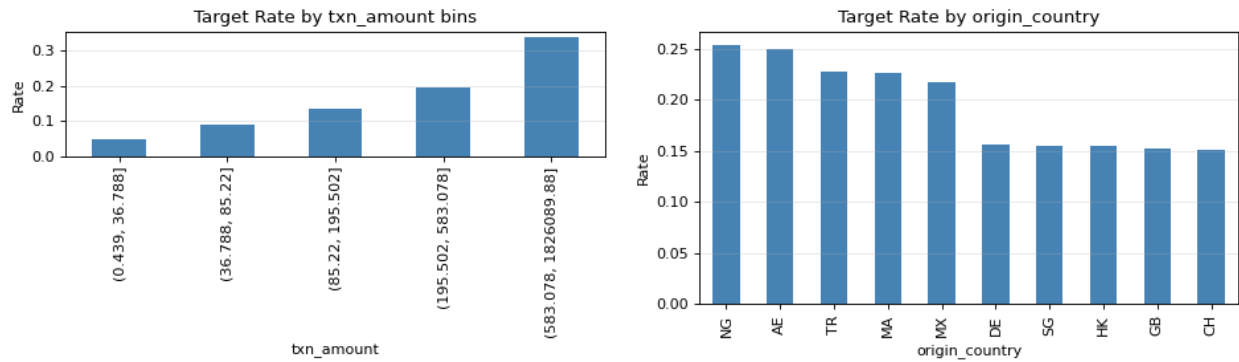
Categorical vs Target

Column	Category	Target Rate
origin_country	NG	25.43%
origin_country	AE	24.98%
origin_country	TR	22.76%
origin_country	MA	22.61%
origin_country	MX	21.71%
origin_country	DE	15.60%
origin_country	SG	15.54%
origin_country	HK	15.46%
origin_country	GB	15.20%
origin_country	CH	15.06%
destination_country	NG	36.13%
destination_country	TR	33.49%
destination_country	MX	32.74%
destination_country	MA	32.32%
destination_country	AE	32.14%
destination_country	SG	8.45%
destination_country	IN	8.30%
destination_country	GB	8.09%
destination_country	IT	7.92%
destination_country	BR	7.44%
ip_country	ES	17.32%
ip_country	AE	17.03%
ip_country	CN	16.72%
ip_country	DE	16.63%
ip_country	SG	16.45%
ip_country	MA	16.38%
ip_country	US	16.16%
ip_country	IN	16.08%
ip_country	HK	15.97%
ip_country	FR	15.74%
currency	NGN	25.43%
currency	AED	24.98%
currency	TRY	22.76%
currency	MAD	22.61%

Column	Category	Target Rate
currency	MXN	21.71%
currency	SGD	15.54%
currency	HKD	15.46%
currency	GBP	15.20%
currency	CHF	15.06%
currency	EUR	15.04%
payment_rail	swift	31.48%
payment_rail	wire	29.40%
payment_rail	zelle	24.11%
payment_rail	cashapp	23.86%
payment_rail	paypal	13.00%
payment_rail	google_pay	12.45%
payment_rail	stripe	12.16%
payment_rail	adyen	11.71%
payment_rail	apple_pay	9.40%
payment_rail	venmo	9.38%
merchant_category	luxury	19.35%
merchant_category	travel	18.56%
merchant_category	charity	16.09%
merchant_category	grocery	15.84%
merchant_category	electronics	15.69%
merchant_category	utilities	15.65%
merchant_category	restaurants	15.51%
merchant_category	gaming	15.11%
merchant_category	rideshare	15.10%
merchant_category	crypto	15.01%
txn_type	international_remit	35.12%
txn_type	crypto_exchange	33.40%
txn_type	cash_withdrawal	21.30%
txn_type	p2p_transfer	9.18%
txn_type	bill_pay	7.81%
txn_type	merchant_pay	7.63%
txn_type	loan_payment	7.35%
txn_type	cash_deposit	6.93%
txn_channel	api	17.61%
txn_channel	web	16.12%
txn_channel	mobile_app	15.77%
txn_channel	branch	15.31%
txn_channel	call_center	15.22%
device_type	UNKNOWN	17.03%
device_type	ios	16.25%
device_type	android	16.10%
device_type	web	15.86%

Column	Category	Target Rate
device_type	api	15.12%
is_pep	1	49.55%
is_pep	0	14.43%

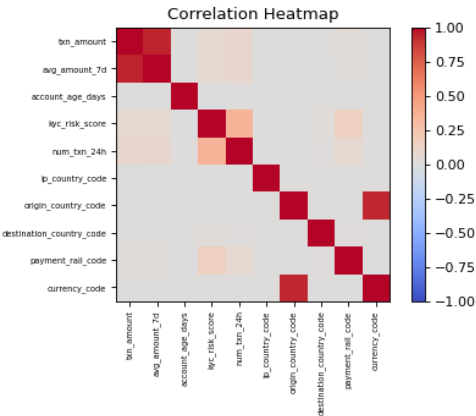
Charts



Feature Vs Feature

- No highly correlated feature pairs detected.

Charts



Time Drift

- Time series and drift analysis completed.

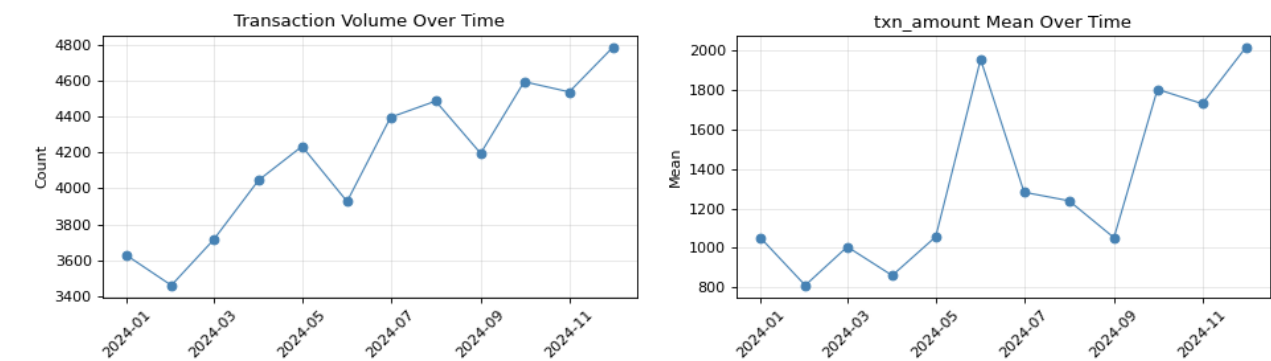
PSI Drift (Numeric)

Column	PSI
payment_rail_code	0.290
num_txn_24h	0.281
destination_country_code	0.185
kyc_risk_score	0.131
txn_amount	0.044
avg_amount_7d	0.036
account_age_days	0.000
ip_country_code	0.000
origin_country_code	0.000
currency_code	0.000

Categorical Drift (Total Variation)

Column	Drift Score
txn_type	0.292
payment_rail	0.265
destination_country	0.168
is_pep	0.037
merchant_category	0.014
currency	0.010
origin_country	0.009
txn_channel	0.008
ip_country	0.007
device_type	0.005

Charts



Skipped EDA Sections

No sections were skipped.

Run Configuration

Key	Value
data_path	bank_aml_tm_synthetic_50k_local=data/bank_aml_tm_synthetic_50k_local.csv
rows_original	50,000
rows_used	50,000
target_col	sar_label
time_col	value_date
time_parse_ratio	1
composition	{'mode': 'single_table', 'base_table': 'bank_aml_tm_synthetic_50k_local', 'tables': ['bank_aml_tm_synthetic_50k_local']}